Feature Selection

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The importance of features (attributes) selection

- Reduce the cost of learning by reducing the number of attributes.
- Provide **better learning performance** compared to using the full attribute set.
- Reduce the cost of feature measurement.

Methods

There are two approach for attribute selection.

1. Filter approach attempt to assess the merits of attributes from the data, ignoring the classifier.
2. Wrapper approach the attributes subset selection is done using the classifier as a black box.

Ron Kohavi, George H. John (1997). Wrappers for feature subset selection. Artificial Intelligence. 97(1-2):273-324.

Filters

- Given *n* original features, how do you select size of subset
  - User can preselect a size *p* ($< n$) – not usually as effective
  - Can find the smallest size where adding more features does not yield improvement
- Filters work independent of any particular learning algorithm
- Filters seek a subset of features which maximize some type of between class separability – or other merit score
- Can score each feature independently and keep best subset
  - e.g. 1^{st} order correlation with output, fast, less optimal
- Can score subsets of features together
  - Exponential number of subsets requires a more efficient, sub-optimal search approach

Info Gain

- Information Gain (IG) is a crucial metric to determine the relevance or importance of a particular feature in predicting or classifying a target variable.
- It essentially quantifies the reduction in uncertainty (measured by entropy) about the target variable when we know the value of a specific attribute.
- Information gain measures how much "information" a feature provides about the class label. A higher information gain signifies that a feature is more effective in separating the data into distinct classes, thus reducing the uncertainty about which class a particular instance belongs to.

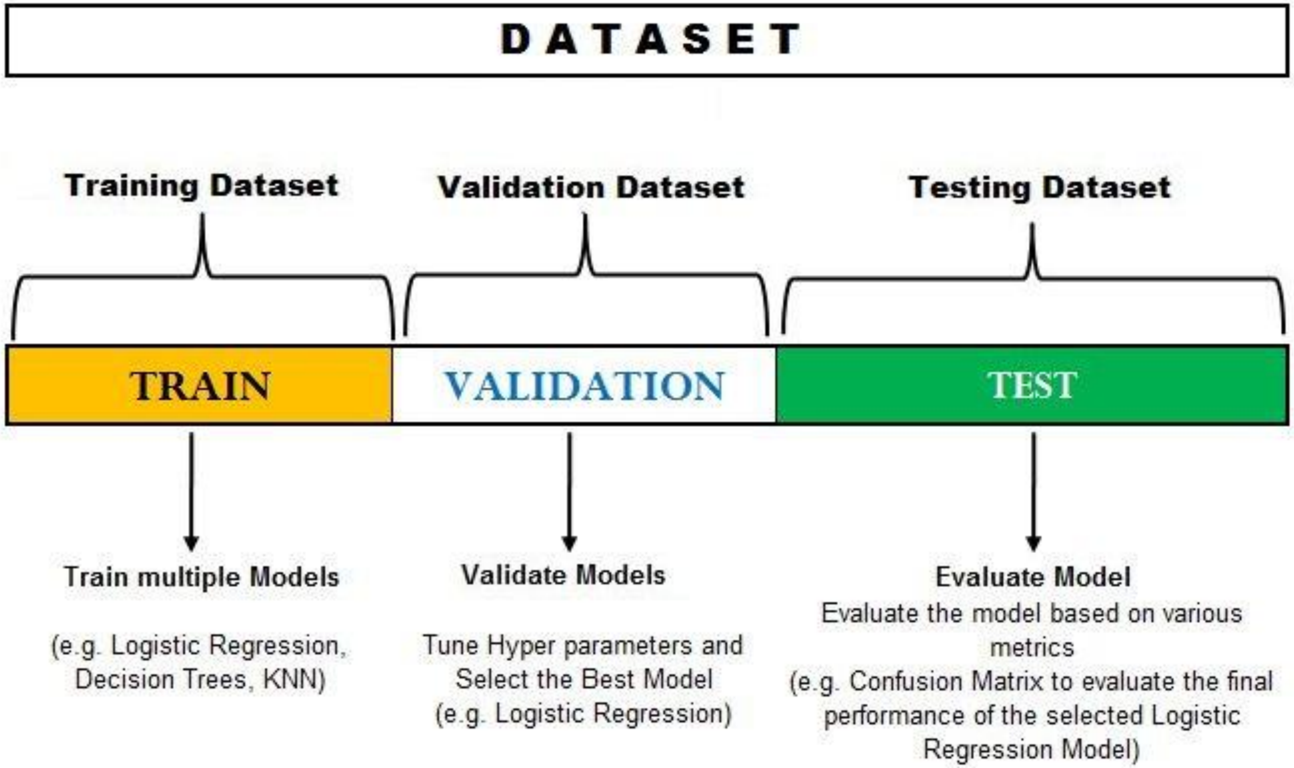
Spearman Correlation

- Spearman correlation, also known as Spearman's rank correlation coefficient or Spearman's rho, is a non-parametric measure of the strength and direction of the association between two ranked variables.
- Unlike Pearson correlation, which assesses linear relationships between raw data, Spearman correlation evaluates the monotonic relationship between the rankings of the data points.
- A monotonic relationship exists when one variable tends to increase as the other increases (or tends to decrease as the other increases), but not necessarily at a constant rate.
- The Spearman correlation coefficient ranges from -1 to +1.

Feature Selection - Wrappers

- Optimizes for a specific learning algorithm
- The feature subset selection algorithm is a "wrapper" around the learning algorithm
    1. Pick a feature subset and pass it in to learning algorithm
    2. Create training/test set based on the feature subset
    3. Train the learning algorithm with the training set
    4. Find accuracy (objective) with validation set
    5. Repeat for all feature subsets and pick the feature subset which led to the highest predictive accuracy (or other objective)
- Basic approach is simple
- Variations are based on how to select the feature subsets, since there are an exponential number of subsets

TUNING (WRAPPER ETC.)



Feature Selection - Wrappers

- Exhaustive Search - Exhausting
- Forward Search – O(n^2 · learning/testing time) - Greedy
  1. Score each feature by itself and add the best feature to the initially empty set FS (FS will be our final Feature Set)
  2. Try each subset consisting of the current FS plus one remaining feature and add the best feature to FS
  3. Continue until stop getting significant improvement (over a window)
- Backward Search – O(n^2 · learning/testing time) - Greedy
  1. Score the initial complete set FS (FS will be our final Feature Set)
  2. Try each subset consisting of the current FS minus one feature in FS and drop the feature from FS causing least decrease in accuracy
  3. Continue until begin to get significant decreases in accuracy
- Other heuristic approaches available…

Conclusions

- Filter approaches are very fast and are particularly useful if number of attributes is large.
- Wrapper algorithms require extensive computation, and are particularly useful for improving the accuracy of a specific classifiers.

LAB

- Compare the accuracy with/without feature selection
  - Take one dataset.
  - Take three classifiers: RandomForest / NaiveBayes / Ibk
  - Perform 10-fold cross validation
- Is the evaluation fair?
- API or GUI?